

Calibration, Registration and Navigation with a Distorted EM Tracker

Programming Assignments 1 and 2, EN.601.455/655 Computer Integrated Surgery I (Johns Hopkins)

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1 Introduction

This report describes my implementation of Programming Assignments 1 and 2 of CIS I. A simulated stereotactic navigation system has an electromagnetic (EM) tracker with a large, repeatable distortion and some noise, an accurate optical tracker that jiggles on its tripod, a calibration object carrying both EM markers and optical LEDs, a dimpled post, and two pointer probes (one EM, one optical). PA1 builds the basic tools (frames, point-set registration, pivot calibration) and computes where the EM markers *should* have been measured. PA2 uses those expected positions to fit and remove the EM distortion, then registers the EM tracker to a CT image and reports the probe tip in CT coordinates.

The code is in Python (package `cisnav`), uses NumPy and SciPy only for linear algebra, and is checked by 506 automated tests and by comparison with every debug data set. The registration and calibration routines are written from scratch, as the course rules require.

2 Mathematical approach

Notation. A frame $F = [R, \vec{p}]$ maps a point $\vec{b}$ to $F\vec{b} = R\vec{b} + \vec{p}$, with R a rotation ($R^T R = I$, $\det R = +1$). Frames compose as $F_1 F_2 = [R_1 R_2, R_1 \vec{p}_2 + \vec{p}_1]$ and invert as $F^{-1} = [R^T, -R^T \vec{p}]$. F_{AB} maps coordinates in frame B into frame A .

2.1 Point-set to point-set registration

Given corresponding points $\vec{a}_i$ and $\vec{b}_i$, $i = 1..N$, find F minimising $\sum_i \|R\vec{a}_i + \vec{p} - \vec{b}_i\|^2$. I use the SVD method of Arun, Huang and Blostein (1987) with the reflection correction of Umeyama (1991).

1. *Translation.* Setting the derivative with respect to $\vec{p}$ to zero gives $\vec{p} = \bar{\vec{b}} - R\bar{\vec{a}}$, where $\bar{\vec{a}}, \bar{\vec{b}}$ are the centroids. With centred points $\tilde{\vec{a}}_i = \vec{a}_i - \bar{\vec{a}}$ and $\tilde{\vec{b}}_i = \vec{b}_i - \bar{\vec{b}}$ only the rotation is left.
2. *Rotation.* Expanding the squared norm, $\sum \|R\tilde{\vec{a}}_i - \tilde{\vec{b}}_i\|^2 = \sum \|\tilde{\vec{a}}_i\|^2 + \sum \|\tilde{\vec{b}}_i\|^2 - 2 \operatorname{tr}(RH)$ with $H = \sum_i \tilde{\vec{a}}_i \tilde{\vec{b}}_i^T$. Minimising the error is maximising $\operatorname{tr}(RH)$.
3. *SVD.* Write $H = USV^T$. Then $\operatorname{tr}(RH) = \operatorname{tr}(MS)$ with $M = V^T R U$ orthogonal, and $\operatorname{tr}(MS) = \sum_k M_{kk} s_k \leq \sum_k s_k$, with equality at $M = I$, that is $R = VU^T$.
4. *Reflection.* If $\det(VU^T) = -1$ (possible with noisy or planar data), that matrix is a reflection. The best proper rotation flips the term with the smallest singular value, which costs least: $R = V \operatorname{diag}(1, 1, \det(VU^T)) U^T$.

The routine rejects fewer than 3 points and collinear or coincident sets (second singular value near zero), where the rotation about the line is undetermined. Horn's (1987) unit-quaternion method solves the same problem and would give the same answer; I implemented only the SVD form.

2.2 Pivot calibration

While the probe tip sits in a dimple and the body is swung around it, the tracker reports poses $F_k = [R_k, \vec{p}_k]$, $k = 1..K$. The tip offset $\vec{t}$ (probe coordinates) and the dimple $\vec{P}$ (tracker coordinates) are fixed, so every frame satisfies $R_k \vec{t} + \vec{p}_k = \vec{P}$, or

$$\begin{bmatrix} R_k & -I \end{bmatrix} \begin{bmatrix} \vec{t} \\ \vec{P} \end{bmatrix} = -\vec{p}_k.$$

Stacking all frames gives a $3K \times 6$ linear system that I solve by linear least squares (`numpy.linalg.lstsq`). The poses come from registration, $F_k = \text{register}(\vec{g}, \vec{G}[k])$, against a centred probe model $\vec{g}$. The handout suggests the first frame's markers minus their centroid, $\vec{g}_j = \vec{G}_j[1] - \vec{G}[1]$. That is exact for a rigid probe, but distorted EM readings change the marker shape from frame to frame, and the post then depends on which frame happens to come first. I use the generalized Procrustes mean instead: align every frame to the current estimate, average, recentre and repeat, starting from the first frame. For rigid readings this gives the same $\vec{g}$; for distorted ones every frame counts equally and the result no longer depends on the frame order (a unit test checks both). The system has full rank only if the probe rotates about at least two different axes. With rotation about a single axis the tip and post offsets along that axis cannot be separated (a unit test checks this rank loss).

2.3 Expected EM marker positions (PA1)

For each calibration frame, $F_D = \text{register}(\vec{d}, \vec{D})$ is the pose of the EM base in optical tracker coordinates and $F_A = \text{register}(\vec{a}, \vec{A})$ the pose of the calibration object. Chaining them gives the object pose in EM coordinates, and so where an ideal EM tracker would see the EM markers: $\vec{C}_i^{exp} = F_D^{-1} F_A \vec{c}_i$.

For the optical probe, the optical tracker may move between frames, so each frame's readings are first expressed in EM coordinates with that frame's own F_D : $\vec{H}_j^{EM} = F_D^{-1} \vec{H}_j$. Pivot calibration on $\vec{H}^{EM}$ then gives the post in EM coordinates.

2.4 Distortion correction (PA2)

The EM distortion is smooth and repeatable, so it is modelled by a polynomial P that maps measured positions $\vec{q}$ to true positions: $P(\vec{C}) \approx \vec{C}^{exp}$.

- *Scaling.* Each coordinate is mapped to $[0, 1]$ with the bounding box of the calibration readings: $u = (q - q_{min}) / (q_{max} - q_{min})$. The box does not change the fitted function (an affine change of variables maps degree- N polynomials onto themselves); it only keeps the basis well conditioned.
- *Basis.* The Bernstein polynomials $B_{N,k}(u) = \binom{N}{k} u^k (1-u)^{N-k}$, combined as the tensor product $F_{ijk}(\vec{u}) = B_{N,i}(u_x) B_{N,j}(u_y) B_{N,k}(u_z)$, give $(N+1)^3$ basis functions.
- *Fit.* Each of the $125 \times 27 = 3375$ calibration points gives one row of basis values; the coefficient matrix c ($((N+1)^3 \times 3)$) solves $Fc \approx C^{exp}$ in the least-squares sense, one `lstsq` call for all three output coordinates.
- *Degree.* Too low a degree cannot follow the distortion; too high a degree fits noise and oscillates between samples and near the edges. I choose N by 5-fold cross-validation, holding out whole calibration frames, and take the smallest degree within 5% of the best held-out error. This uses only the calibration data, never the answer files.

2.5 Registration to CT and navigation (PA2)

After dewarping the pivot readings and repeating the pivot calibration, each fiducial frame gives the tip in EM coordinates, $\vec{B}_j = F_G[j] \vec{t}$, where $F_G[j]$ is the registration of the probe model to the dewarped readings. Then $F_{reg} = \text{register}(\vec{B}, \vec{b})$ maps EM to CT. For each navigation frame the tip in CT is $\vec{v}_k = F_{reg} F_G[k] \vec{t}$.

3 Algorithm steps

PA1, for each data set:

1. Read the calibration body, calibration readings, EM pivot and optical pivot files.
2. For every calibration frame compute F_D , F_A and $\vec{C}^{exp} = F_D^{-1} F_A \vec{c}$.
3. EM pivot: define $\vec{g}$ from frame 1, register every frame, solve the stacked system for $\vec{t}_G$ and $\vec{P}$.
4. Optical pivot: map $\vec{H}$ into EM coordinates with each frame's F_D , then pivot as in step 3.
5. Write NAME-output1.txt: header, EM post, optical post, all $\vec{C}^{exp}$.

PA2, for each data set:

1. Compute $\vec{C}^{exp}$ for all 125 calibration frames (PA1 step 2).
2. Choose the degree by cross-validation, then fit the Bernstein correction P to all frames.
3. Dewarp the EM pivot readings with P and repeat the pivot calibration.
4. Dewarp the fiducial readings and compute the tip positions $\vec{B}_j$.
5. Compute F_{reg} from $\vec{B}_j$ and the CT fiducials $\vec{b}_j$.
6. Dewarp each navigation frame, compute the tip, map it with F_{reg} and write NAME-output2.txt. A PA2 output1 with the dewarped EM post is also written when the optical pivot file exists.

4 Program overview

Module	Purpose
cisnav/io.py	Readers for every input file; header-tolerant, checks row counts
cisnav/frames.py	Rotations (axis-angle, quaternion) and the Frame class
cisnav/registration.py	SVD point-set registration with reflection fix
cisnav/pivot.py	Stacked least-squares pivot calibration
cisnav/distortion.py	Bernstein basis, correction fit, cross-validation
cisnav/pa1.py	PA1 steps and command line tool
cisnav/pa2.py	PA2 steps and command line tool
cisnav/output.py	Output writers in the handout format
scripts/compare_debug.py	Validation against the debug outputs
scripts/error_sources.py,	Error analysis and figures
scripts/monte_carlo.py	

The modules form layers: **frames** has no dependencies, **registration** uses **frames**, **pivot** uses **registration**, and the **pa1** and **pa2** pipelines only combine these building blocks with the readers and writers. Every step is a small function with a docstring that states its equation, so each can be tested on its own. Usage:

```
pip install -e ".[dev]"
python -m cisnav.pa1 --data-dir data/pa1 --set all --out output/
python -m cisnav.pa2 --data-dir data/pa2 --set all --out output/
python scripts/compare_debug.py --assignment pa2
```

Libraries: NumPy (Harris et al. 2020) for arrays, SVD (**linalg.svd**) and least squares (**linalg.lstsq**), both backed by LAPACK; SciPy (Virtanen et al. 2020) for binomial coefficients; Matplotlib (Hunter 2007) for figures; pytest for tests. No registration or calibration library was used.

5 Validation

Validation has two layers.

Unit tests on synthetic data with known answers (506 tests, run on every push by GitHub Actions). Examples: exact recovery of random frames by registration, including planar point sets and a mirrored data set that tempts the unconstrained SVD into a reflection; least-squares optimality (any small rotation of the solution increases the residual); residual size under noise matching the expected $\sigma\sqrt{(3N-6)/N}$; exact pivot recovery

and rank loss under single-axis rotation; exact recovery of a cubic correction by a degree-3 fit and sub-0.01 mm correction of a smooth non-polynomial warp on held-out points; cross-validation recovering the true degree; and readers loading all 132 data files with counts checked against their headers.

Debug sets. Each debug set switches on one error source, so a mismatch points to its cause. All values below are 3D distances in mm; both our files and the reference files are rounded to 0.01 mm.

PA1 set	Sources on	C^{exp} max	C^{exp} RMS	EM post	Optical post
a	none	0.017	0.005	0.000	0.000
b	noise	0.751	0.491	0.000	0.000
c	distortion	0.953	0.403	0.000	0.000
d	jiggle	0.024	0.012	0.000	0.000
e	distortion, jiggle	3.741	1.710	0.010	0.000
f	all three	4.282	1.788	0.010	0.010
g	all three	3.359	1.680	0.010	0.000

Both pivot calibrations match all seven sets to rounding. C^{exp} matches in a and d, where the EM tracker has no error. In the other sets the reference C^{exp} itself contains EM-side error: in set b it is identical, digit for digit, to the measured noisy EM readings, and in c, e, f and g it is not a rigid image of the calibration body (rigid-fit residual 0.46 to 2.1 mm per frame, against 0.005 mm for ours). Since $F_D^{-1}F_A\vec{c}$ depends only on optical data, which has no EM error, I did not change the program to chase these values. The analysis is in `results/pa1_validation.md`.

PA2 set	Sources on	Degree	FRE RMS	Tip max	Tip RMS	Dewarped EM post
a	none	1	0.006	0.014	0.009	0.004
b	noise	1	0.420	0.065	0.048	0.176
c	distortion	4	0.016	0.017	0.013	0.004
d	jiggle	1	0.007	0.014	0.010	0.003
e	all three	4	0.204	0.197	0.154	0.006
f	all three	4	0.329	0.064	0.054	0.007

Navigation matches to rounding in a, c and d. Set c is the real test of the distortion correction: the raw EM pivot residual there is 1.2 mm and the dewarped one 0.03 mm. In set b the dewarped post differs by 0.18 mm; forcing degree 4 reproduces the reference to 0.006 mm, so the reference most likely used degree 4 everywhere. Set b has no distortion and noise-free pivot readings, so its plain pivot is the true post; our degree-1 result is 0.03 mm from it, the reference 0.18 mm. In e and f the tip differences (up to 0.2 mm) remain although the degree and the post agree with the reference, and are the size of the EM noise effect on single fiducial and navigation readings (FRE 0.20 and 0.33 mm). Details are in `results/pa2_validation.md`.

6 Results for the unknown data sets

PA1 (EM coordinates, mm). The pivot residual is the RMS of $\|R_k\vec{t} + \vec{p}_k - \vec{P}\|$ over frames.

Set	EM post	EM pivot RMS	Optical post	Optical pivot RMS
h	209.47, 195.41, 217.00	6.21	394.59, 399.97, 192.83	0.008
i	206.29, 200.55, 194.22	2.55	404.07, 398.23, 203.91	0.008
j	191.07, 190.53, 210.24	1.06	397.66, 408.18, 202.79	0.004
k	191.10, 201.09, 187.52	2.08	402.19, 403.11, 197.89	0.007

PA2. Degree and held-out calibration error from cross-validation, pivot RMS after dewarping, FRE of F_{reg} and the dewarped EM post (mm):

Set	Degree	CV error	Pivot RMS	FRE	Dewarped EM post
g	4	0.405	0.084	0.045	204.84, 205.46, 190.78
h	4	0.388	0.103	0.177	193.43, 209.38, 200.68
i	4	0.408	0.140	0.136	195.80, 210.00, 192.44
j	4	0.438	0.089	0.238	192.21, 196.20, 192.79

Probe tip positions in CT coordinates (mm), the contents of `output2`:

Frame	g	h
1	111.86, 75.83, 148.57	116.55, 63.62, 146.84
2	85.56, 143.10, 75.27	42.11, 164.77, 72.88
3	76.58, 47.66, 92.43	66.43, 107.50, 61.56
4	65.91, 80.82, 26.43	41.37, 140.69, 26.82

Frame	i	j
1	63.74, 91.59, 108.97	143.12, 114.10, 59.86
2	106.37, 165.62, 42.22	96.88, 33.88, 114.63
3	73.77, 145.75, 122.45	134.03, 41.27, 109.16
4	50.04, 87.77, 88.77	126.36, 46.95, 117.73

7 Discussion

What each error source costs. Figure 1 runs the PA2 pipeline on every debug set with and without the correction. *Jiggle* (set d) costs nothing, because every optical reading is mapped through its own frame’s F_D . *Noise* (set b) sets a floor that no correction removes: about 0.5 mm RMS in the calibration readings and 0.42 mm FRE. *Distortion* dominates: in set c it raises the navigation error to 0.93 mm, and in set e, combined with the other sources, to 7.3 mm. The correction brings these to 0.016 mm and 0.20 mm. The unknown sets behave like the worst debug sets (raw calibration error 5.3 to 6.3 mm, held-out error after correction about 0.4 mm), and their FRE values (0.05 to 0.24 mm) are similar to sets e and f, so I expect their navigation errors to be a few tenths of a millimetre.

Polynomial degree. Figure 2 shows the held-out error against degree. In every distorted set it falls by roughly an order of magnitude between degree 3 and 4, then rises: at degree 6 it is 6 to 60 times the degree-4 value, and at degree 7 (512 coefficients from about 2700 training points) the fits oscillate wildly. The jump at 4 suggests that the simulated distortion is close to a polynomial of that degree. Degree 1 wins in the sets without distortion, which avoids fitting noise (the effect seen in set b).

Where the correction is weakest. Figure 3 shows the measured EM error in a slice of the workspace and what is left after correction, on frames that were held out of the fit. The 9 mm RMS error shrinks to 0.6 mm, but the largest remaining errors sit at the edge of the volume ($x \approx 870$ mm), where the polynomial is least constrained. Between 7% and 19% of the pivot readings lie slightly outside the calibration box, so the polynomial extrapolates there; the fiducial and navigation readings are all inside it.

Sensitivity to noise (Monte Carlo). Figure 4 registers the real marker geometries under synthetic noise. The error grows linearly with noise, and more widely spread markers give smaller rotation errors (27 body markers beat the 6 probe markers by a factor of 11). The probe-tip error, 100 mm from the marker centroid, agrees within 4% with the prediction of Fitzpatrick, West and Maurer (1998), $\text{TRE}^2 = \frac{\text{FLE}^2}{N} (1 + \frac{1}{3} \sum_k d_k^2 / f_k^2)$, which is an independent check of the registration code. Figure 5 does the same for pivot calibration: error falls

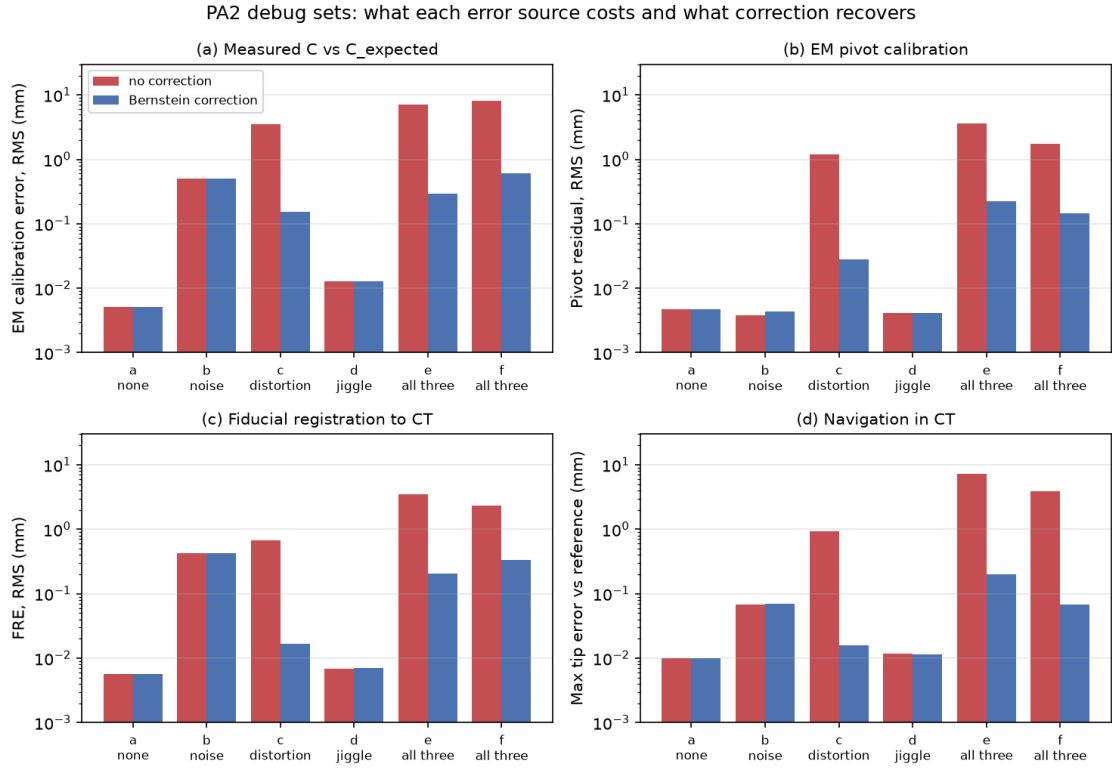


Figure 1: PA2 debug sets with and without distortion correction (log scale).

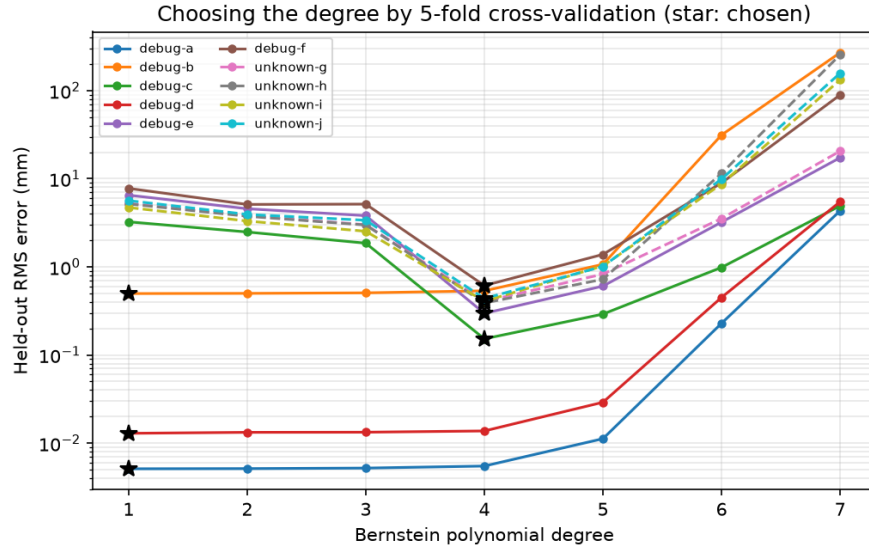


Figure 2: Held-out correction error against Bernstein degree for all PA2 sets.

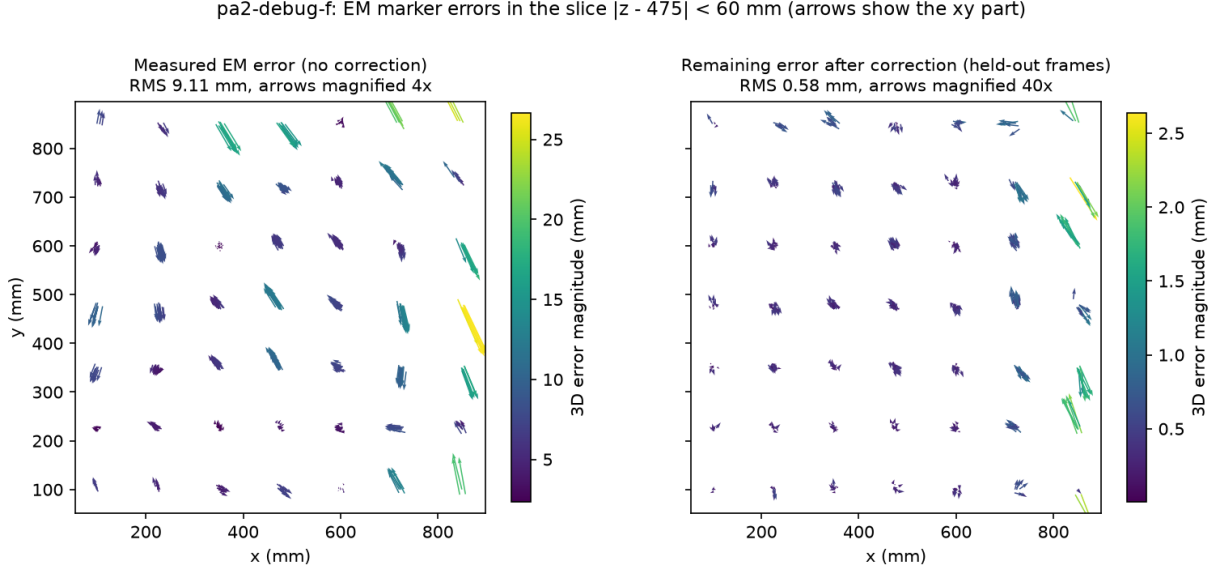


Figure 3: Distortion in a slice of pa2-debug-f before and after correction (held-out frames).

with the number of frames (from 0.72 mm with 6 frames to 0.19 mm with 48, at $\sigma = 0.3$ mm), and rises steeply when the probe is tilted only a little (3.9 mm at 2 degrees against 0.42 mm at 30 degrees). Small tilts make the system nearly rank deficient, as the theory predicts.

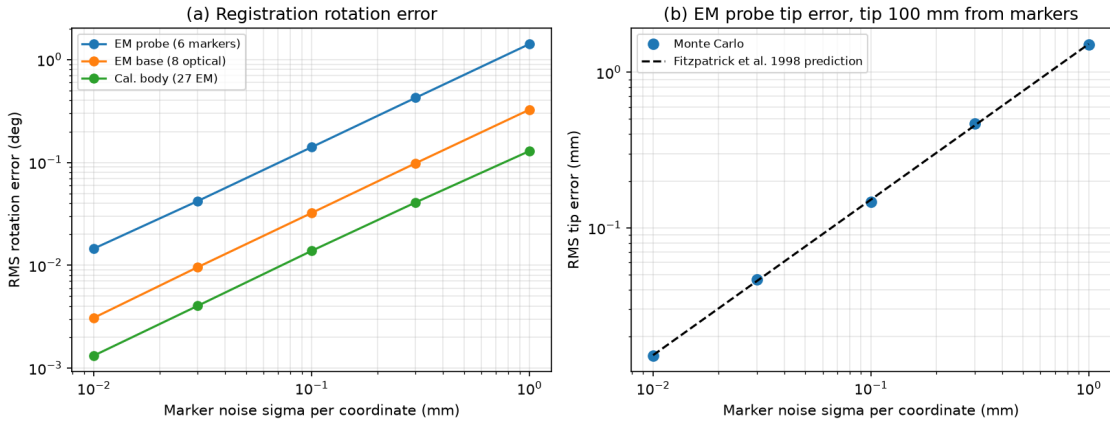


Figure 4: Registration error against marker noise; tip error compared with the Fitzpatrick prediction.

Limitations. The correction assumes the optical tracker is exact; any optical error would be learned as distortion. The degree is chosen per data set, which is principled but means sets can differ in model complexity (it differs from the reference only in set b, where our choice is the more accurate one). The PA1 EM posts for the unknown sets are not distortion corrected: their pivot residuals of 1 to 6 mm show that they carry a distortion bias of the same order, which PA2 removes. Finally, I could not explain how the reference C^{exp} files acquire their EM-side error; I tested and rejected the idea that they are polynomial-warped readings.

8 Who did what

This is an individual project by Parmida Mazloomi.

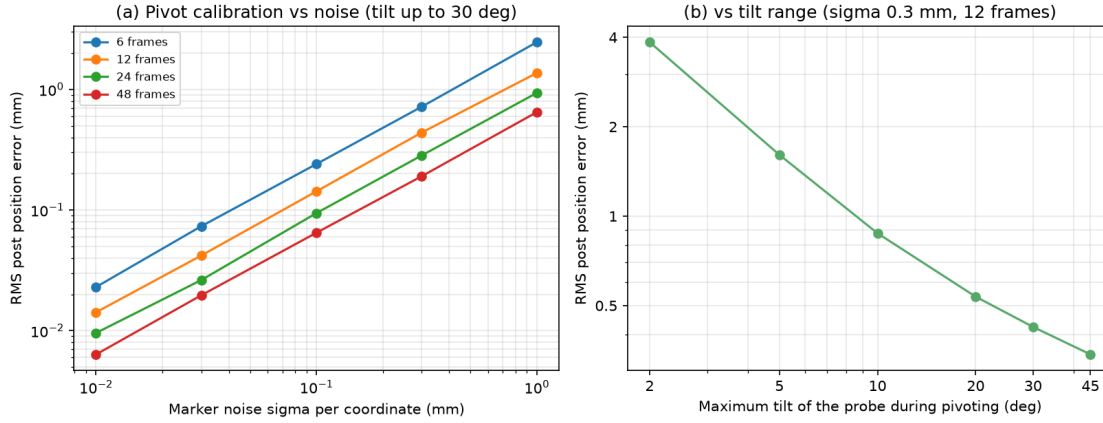


Figure 5: Pivot calibration post error against noise, number of frames and tilt range.

9 References

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